

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2338

Tyson's 138 kV
November 30, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2338. The Interconnection Customer proposes a 183.1 MW solar photovoltaic facility interconnecting at the Tyson's 138 kV substation in the Virginia area. The request was received on August 06, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 540 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$965,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Still Mill-Hunter 138kV Line	Comanche Peak 345/1 kV Transformer Bank	104.7	212	20.7
Tyson's-Hunter 138kV L	Comanche Peak 345/1kV Auto	101.2	289	20.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Still Mill-Hunter 138kV Line	Reconductor line	\$652,000	50% (\$328,000)
Hunter-Tyson's 138kV	Reconductor line	\$1,267,000	50% (\$637,000)
	Total allocated to project		\$965,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.